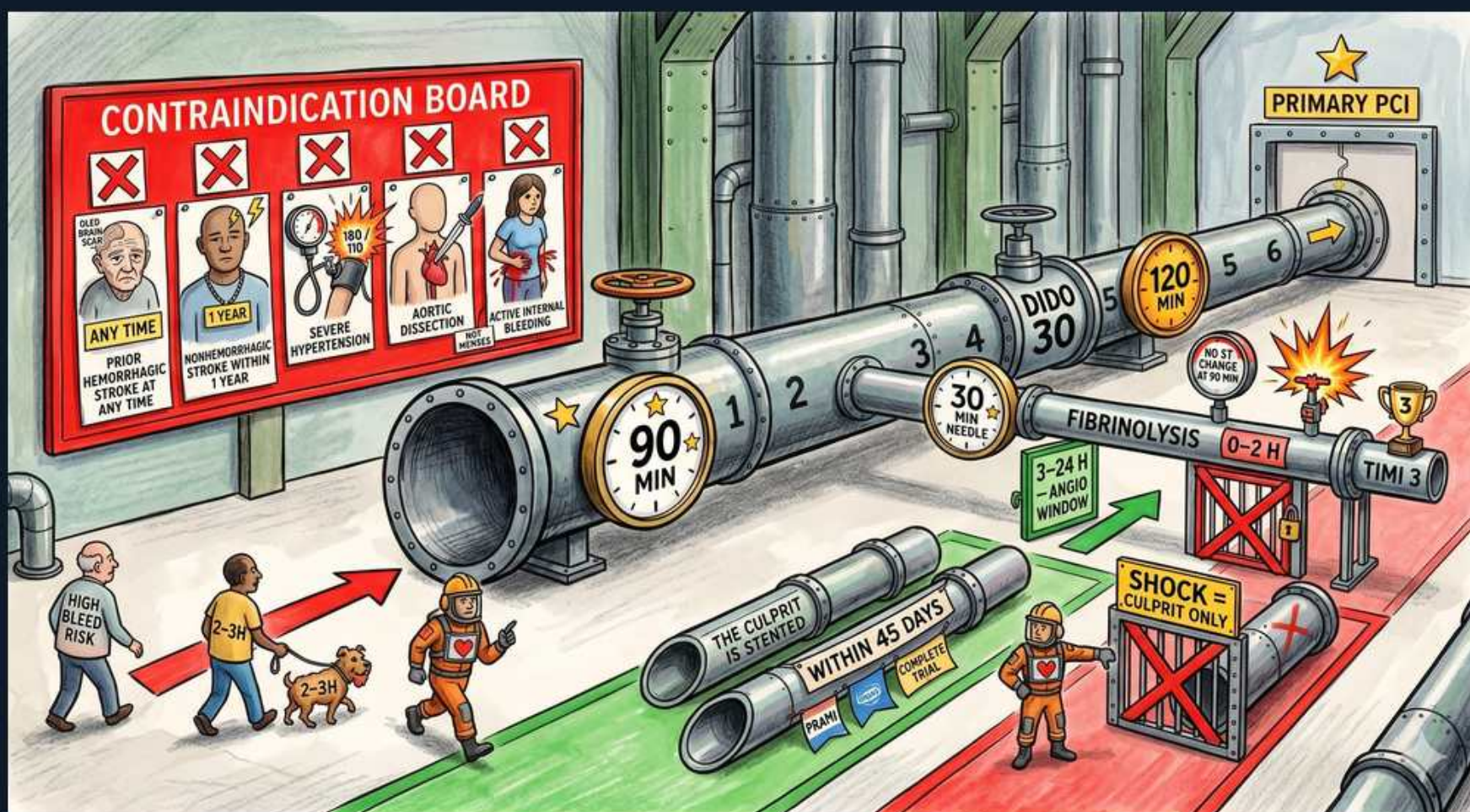


IHD / STEMI — reperfusion (PCI vs fibrinolysis)



1. Primary PCI gold star

WHAT YOU SEE

Gold 'PRIMARY PCI' star top-right

WHAT IT ENCODES

Primary PCI is the preferred reperfusion strategy for STEMI whenever it can be delivered in a timely fashion at a PCI-capable center. It restores flow more reliably than lytics, has lower rates of death, reinfarction, and intracranial hemorrhage, and lets you mechanically open the culprit lesion. Indications: any STEMI (ST elevation or new LBBB) within 12h of symptom onset, and still beneficial up to 12–24h if ongoing ischemia, shock, or hemodynamic instability. Mnemonic: 'balloon beats clot-buster when the cath lab is close.'

BOARD TRAP

WRONG: choosing fibrinolysis over PCI when a patient presents to a PCI-capable hospital. Discriminator: if the cath lab is available and door-to-balloon can be met, PCI is always preferred — lytics are the fallback only when PCI cannot be delivered in time.

2. Door-to-balloon 90 MIN

WHAT YOU SEE

Big '90 MIN' clock on pipe

WHAT IT ENCODES

At a PCI-capable hospital the target is door-to-balloon (first medical contact to device) ≤90 minutes. This is the benchmark quality metric for primary PCI and every minute of delay increases infarct size ('time is muscle'). The fastest reperfusion correlates with the best myocardial salvage and survival.

BOARD TRAP

WRONG answer: stating the door-to-balloon goal is 120 minutes. Discriminator: 90 min is the in-house PCI target; 120 min is the OUTSIDE limit of total time-to-PCI (including transfer) before you should give lytics instead — don't confuse the two thresholds.

3. DIDO 30

WHAT YOU SEE

'DIDO 30' valve

WHAT IT ENCODES

Door-in–door-out (DIDO) target is ≤30 minutes: at a non-PCI hospital, the patient should be diagnosed and shipped out within 30 min so the total transfer-to-balloon time can still meet ≤120 min. A DIDO >30 min predicts worse outcomes and may tip the decision toward giving lytics locally.

BOARD TRAP

WRONG: DIDO target is 90 minutes. Discriminator: 30 min is for getting the patient back OUT the door of a transferring hospital; confusing it with the 90-min in-house balloon time is the classic mix-up.

4. PCI within 120 MIN

WHAT YOU SEE

'120 MIN' wheel near transfer arrow

WHAT IT ENCODES

If the patient is at a non-PCI center, transfer for primary PCI is preferred only if total first-medical-contact-to-device time is ≤120 minutes. If PCI cannot be achieved within 120 min, give fibrinolysis first (within 30-min door-to-needle) then transfer. This is the single most tested branch point in STEMI triage.

BOARD TRAP

WRONG: transfer everyone for PCI regardless of how long it takes. Discriminator: beyond an anticipated 120 min to balloon, lytics save more muscle than waiting for a delayed PCI — choose pharmacologic reperfusion, not endless transfer.

5. Fibrinolysis door-to-needle 30

WHAT YOU SEE

'30 NEEDLE' valve, 'FIBRINOLYSIS'

WHAT IT ENCODES

When lysis is the strategy, door-to-needle should be ≤30 minutes. Preferred agents are fibrin-specific lytics: tenecteplase (TNK, single weight-based IV bolus — easiest), alteplase (tPA, bolus + infusion), or reteplase. Give with adjunctive aspirin, a P2Y12 inhibitor (clopidogrel), and anticoagulation (enoxaparin or UFH). Always pair with a plan for routine angiography 3–24h later (pharmacoinvasive strategy).

BOARD TRAP

WRONG: door-to-needle target is 90 minutes. Discriminator: needle time is fast (≤30 min) — the 90-min figure belongs to in-house door-to-balloon, not to lytic administration.

6. 0–2 H / 3–12 H window

WHAT YOU SEE

'0–2 H' and '3–12 H window' green sign

WHAT IT ENCODES

Fibrinolysis is most effective when given early — greatest myocardial salvage in the first 0–2 hours from symptom onset, with declining but still real benefit through ~12 hours. Beyond 12h, lytics offer little benefit and rising harm, so they are generally not given unless ongoing ischemia in a setting where PCI is truly unavailable. The benefit curve is steepest early.

BOARD TRAP

WRONG: fibrinolysis is useful out to 24 hours regardless. Discriminator: the lytic window is roughly ≤12h from symptom onset (best ≤2–3h); after 12h the bleeding risk outweighs benefit — late presenters who still need reperfusion go to PCI, not lysis.

7. Contraindication board

WHAT YOU SEE

Red 'CONTRAINDICATION BOARD' poster

WHAT IT ENCODES

This board lists ABSOLUTE contraindications to FIBRINOLYSIS — they do not apply to primary PCI. This is why PCI is favored in any patient with bleeding risk: PCI has no bleeding-based contraindications of this kind. Whenever a STEMI patient has any item on this board, mechanical reperfusion (PCI) becomes the route of choice.

BOARD TRAP

WRONG: these contraindicate primary PCI. Discriminator: every item here forbids LYTICS only — a patient with active bleeding or prior intracranial hemorrhage should be sent for PCI, not denied reperfusion altogether.

8. Prior hemorrhagic stroke — ANY TIME

WHAT YOU SEE

First panel 'ANY TIME' prior hemorrhagic stroke

WHAT IT ENCODES

Any prior intracranial hemorrhage or hemorrhagic stroke at ANY point in life is an absolute contraindication to fibrinolysis — the risk of recurrent catastrophic intracranial bleed is unacceptable. Also absolute: known cerebral AVM, intracranial malignancy, or structural cerebrovascular lesion. Send these patients for PCI.

BOARD TRAP

WRONG: only a RECENT hemorrhagic stroke matters. Discriminator: for hemorrhagic/intracranial bleed the time window is 'ever' — any history at any time permanently rules out lytics, unlike ischemic stroke which uses a 3-month/1-year cutoff.

9. Ischemic stroke within 1 YEAR

WHAT YOU SEE

Second panel '1 YEAR' ischemic stroke

WHAT IT ENCODES

An ischemic stroke within the recent window (classically within 3 months by ACC/AHA, with caution out to a year) is a contraindication to fibrinolysis. Acute ischemic stroke can hemorrhagically transform under lytic therapy. Older remote ischemic strokes become relative rather than absolute. PCI is the alternative.

BOARD TRAP

WRONG: ischemic stroke is never a contraindication to lysis. Discriminator: a RECENT ischemic stroke is contraindicated; the nuance vs hemorrhagic stroke (any time = absolute) is the favorite distractor pairing.

10. Severe HTN

WHAT YOU SEE

Severe hypertension panel

WHAT IT ENCODES

Severe uncontrolled hypertension (SBP >180 or DBP >110 mmHg) is a contraindication to fibrinolysis because of intracranial hemorrhage risk; it is relative if BP can be lowered before lytic. A history of chronic severe poorly controlled HTN is also a relative contraindication. Control the pressure or choose PCI.

BOARD TRAP

WRONG: blood pressure is irrelevant to lysis decisions. Discriminator: markedly elevated BP at presentation (>180/110) sharply raises bleeding risk — either treat it down first or default to PCI.

11. Aortic dissection

WHAT YOU SEE

Aortic dissection panel

WHAT IT ENCODES

Suspected or known aortic dissection is an absolute contraindication to fibrinolysis — lysing a dissection can cause fatal rupture/tamponade. A type A dissection can also occlude a coronary ostium and mimic STEMI, so always consider it in chest pain with a widened mediastinum, pulse deficit, or new aortic regurgitation before giving lytics.

BOARD TRAP

WRONG: a dissection is safe to lyse. Discriminator: lysing a dissection is lethal — the STEMI may itself be caused by the dissection, so exclude dissection clinically/by imaging before any lytic.

12. Active internal bleeding

WHAT YOU SEE

Active internal bleeding panel

WHAT IT ENCODES

Active internal bleeding or a known bleeding diathesis (excluding menses) is an absolute contraindication to fibrinolysis. Also note recent major surgery/trauma and recent internal bleeding as contraindications. These patients should undergo primary PCI, which carries no such bleeding-based prohibition.

BOARD TRAP

WRONG: an active GI bleed is only a RELATIVE contraindication. Discriminator: active internal bleeding (other than menstruation) is ABSOLUTE — do not lyse; route to PCI instead.

13. TIMI 3 trophy

WHAT YOU SEE

'TIMI 3' trophy on right

WHAT IT ENCODES

The angiographic goal of reperfusion is TIMI 3 (normal, brisk, complete) flow in the culprit artery. The TIMI scale runs 0 (no flow) → 1 (penetration without perfusion) → 2 (partial/delayed) → 3 (normal). Achieving TIMI 3 flow correlates with the best survival; PCI achieves it far more reliably (>90%) than lytics (~50–60%).

BOARD TRAP

WRONG: TIMI 1 flow is an adequate endpoint. Discriminator: only TIMI 3 (full normal flow) is the real success endpoint — TIMI 1–2 means inadequate reperfusion and predicts worse outcomes.

14. Shock = culprit only

WHAT YOU SEE

Red barn 'SHOCK = CULPRIT ONLY'

WHAT IT ENCODES

In STEMI complicated by cardiogenic shock, PCI of the CULPRIT vessel only is favored — the CULPRIT-SHOCK trial showed that immediate routine multivessel PCI increased death/renal failure at 30 days vs culprit-only revascularization. Stage non-culprit lesions later if needed. Emergent revascularization in shock is indicated regardless of time from onset.

BOARD TRAP

WRONG: in shock, perform complete multivessel PCI all at once. Discriminator: CULPRIT-SHOCK established culprit-only PCI as superior in shock; immediate full multivessel intervention worsens outcomes — defer non-culprit lesions.